



Chemistry HL/SL S1.1 Introduction to the particulate nature of matter What is the key difference between a compound and a mixture at the particulate level? IB-CHEMISTRY-S1.1-1 CONCEPT	Chemistry HL/SL S1.1 Introduction to the particulate nature of matter Why can chromatography separate the dyes in a mixture of food colourings? IB-CHEMISTRY-S1.1-2 CONCEPT	Chemistry HL/SL S1.1 Introduction to the particulate nature of matter Which technique is best used to separate a dissolved salt from water to recover the pure liquid solvent? A Distillation B Filtration C Chromatography D Simple decanting IB-CHEMISTRY-S1.1-3 MULTIPLE CHOICE
Chemistry HL/SL S1.1 Introduction to the particulate nature of matter State whether melting of solid wax and burning of a candle wick are physical or chemical changes, and justify each briefly. IB-CHEMISTRY-S1.1-4 PRACTICE	Chemistry HL/SL S1.2 The nuclear atom Define mass number and atomic number, and give the relationship between them and the number of neutrons. IB-CHEMISTRY-S1.2-1 CONCEPT	Chemistry HL/SL S1.2 The nuclear atom In a mass spectrometer, why are heavier ions of the same charge deflected less than lighter ions? IB-CHEMISTRY-S1.2-2 CONCEPT
Chemistry HL/SL S1.2 The nuclear atom An atom of ^{13}C has 6 protons. How many neutrons does it contain? A 6 B 7 C 13 D 19 IB-CHEMISTRY-S1.2-3 MULTIPLE CHOICE	Chemistry HL/SL S1.2 The nuclear atom A sample contains two isotopes of copper: ^{63}Cu (69.2%) and ^{65}Cu (30.8%). Calculate the relative atomic mass of copper from this data. IB-CHEMISTRY-S1.2-4 PRACTICE	Chemistry HL/SL S1.3 Electron configurations State the order in which the 4s and 3d sub-shells are filled and then emptied. IB-CHEMISTRY-S1.3-1 CONCEPT
Chemistry HL/SL S1.3 Electron configurations Why do chromium and copper have anomalous ground-state electron configurations? IB-CHEMISTRY-S1.3-2 CONCEPT	Chemistry HL/SL S1.3 Electron configurations What is the maximum number of electrons that can occupy the 3rd principal energy level, $n = 3$? A 8 B 10 C 18 D 2 IB-CHEMISTRY-S1.3-3 MULTIPLE CHOICE	Chemistry HL/SL S1.3 Electron configurations Write the full electron configuration of a ground-state sulfur atom ($Z = 16$) and state how many unpaired electrons it has. IB-CHEMISTRY-S1.3-4 PRACTICE
Chemistry HL/SL S1.4 Counting particles by mass: the mole Give the equation relating amount of substance (n), mass (m) and molar mass (M), and state the units of each. IB-CHEMISTRY-S1.4-1 CONCEPT	Chemistry HL/SL S1.4 Counting particles by mass: the mole Why might the empirical formula determined experimentally from combustion analysis differ slightly from the true simplest whole-number ratio? IB-CHEMISTRY-S1.4-2 CONCEPT	Chemistry HL/SL S1.4 Counting particles by mass: the mole How many molecules are present in 0.20 mol of CO_2 ? A 6.0×10^{22} B 6.0×10^{23} C 1.2×10^{23} D 1.2×10^{24} IB-CHEMISTRY-S1.4-3 MULTIPLE CHOICE
Chemistry HL/SL S1.4 Counting particles by mass: the mole Calculate the mass of NaOH ($M = 40.0 \text{ g mol}^{-1}$) required to prepare 250 cm^3 of a 0.100 mol dm^{-3} solution. IB-CHEMISTRY-S1.4-4 PRACTICE	Chemistry HL/SL S1.5 Ideal gases (HL) State the ideal gas equation and the units required for each quantity when $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$. IB-CHEMISTRY-S1.5-1 CONCEPT	Chemistry HL/SL S1.5 Ideal gases (HL) Under what conditions does a real gas behave most like an ideal gas, and why? IB-CHEMISTRY-S1.5-2 CONCEPT



Chemistry HL/SL S1.5 Ideal gases (HL) What is the approximate volume occupied by 2.00 mol of an ideal gas at STP, 22.4 dm ³ mol ⁻¹ ? A 24.0 dm ³ B 22.4 dm ³ C 44.8 dm ³ D 11.2 dm ³ IB-CHEMISTRY-S1.5-3 MULTIPLE CHOICE	Chemistry HL/SL S1.5 Ideal gases (HL) Calculate the pressure, in kPa, exerted by 0.100 mol of an ideal gas in a 2.00 dm ³ container at 300 K. IB-CHEMISTRY-S1.5-4 PRACTICE	Chemistry HL/SL S2.1 The ionic model State two factors that increase the lattice enthalpy of an ionic compound. IB-CHEMISTRY-S2.1-1 CONCEPT
Chemistry HL/SL S2.1 The ionic model Why does solid NaCl not conduct electricity, while molten NaCl and aqueous NaCl(aq) both do? IB-CHEMISTRY-S2.1-2 CONCEPT	Chemistry HL/SL S2.1 The ionic model Which compound would be expected to have the highest lattice enthalpy? A Al ₂ O ₃ B MgO C NaCl D KI IB-CHEMISTRY-S2.1-3 MULTIPLE CHOICE	Chemistry HL/SL S2.1 The ionic model State the type of structure formed by ionic compounds and explain, in one sentence, why ionic compounds are typically hard but brittle. IB-CHEMISTRY-S2.1-4 PRACTICE
Chemistry HL/SL S2.2 The covalent model According to VSEPR theory, what determines molecular shape, and what is the effect of a lone pair on bond angle? IB-CHEMISTRY-S2.2-1 CONCEPT	Chemistry HL/SL S2.2 The covalent model Explain why water has a much higher boiling point than methane despite both being small covalent molecules. IB-CHEMISTRY-S2.2-2 CONCEPT	Chemistry HL/SL S2.2 The covalent model What is the molecular shape of SO ₂ , given sulfur has one lone pair and two S=O bonding regions? A Linear B Bent (angular) C Trigonal planar D Tetrahedral IB-CHEMISTRY-S2.2-3 MULTIPLE CHOICE
Chemistry HL/SL S2.2 The covalent model Draw the Lewis structure of CO ₂ (2 C=O double bonds, no lone pairs on C) and state its shape and bond angle. IB-CHEMISTRY-S2.2-4 PRACTICE	Chemistry HL/SL S2.3 The metallic model Describe the metallic bonding model in one sentence. IB-CHEMISTRY-S2.3-1 CONCEPT	Chemistry HL/SL S2.3 The metallic model Why does increasing the number of delocalised electrons per atom generally increase a metal's melting point? IB-CHEMISTRY-S2.3-2 CONCEPT
Chemistry HL/SL S2.3 The metallic model Which statement correctly explains the ductility of metals? A The bonds in metals are directional B Metal cations are held in fixed, rigid positions with no possible movement C Delocalised electrons allow layers of cations to slide without breaking metallic bonds D Metals have no free-moving charge carriers IB-CHEMISTRY-S2.3-3 MULTIPLE CHOICE	Chemistry HL/SL S2.3 The metallic model Explain briefly why metals conduct electricity in both the solid and liquid state, unlike ionic compounds which only conduct when molten or dissolved. IB-CHEMISTRY-S2.3-4 PRACTICE	Chemistry HL/SL S2.4 From models to materials (HL) What structural feature of a polymer chain arrangement (linear vs branched) affects its melting point, and why? IB-CHEMISTRY-S2.4-1 CONCEPT



Chemistry HL/SL S2.4 From models to materials (HL) Why can alloying a pure metal with a different element increase its hardness? IB-CHEMISTRY-S2.4-2 CONCEPT	Chemistry HL/SL S2.4 From models to materials (HL) Which of these is a condensation polymer rather than an addition polymer? A Polyester formed from a diol and a dicarboxylic acid B Poly(propene) formed from propene monomers C Poly(vinyl chloride), PVC, formed from chloroethene monomers D Polytetrafluoroethene (PTFE) formed from tetrafluoroethene monomers IB-CHEMISTRY-S2.4-3 MULTIPLE CHOICE	Chemistry HL/SL S2.4 From models to materials (HL) State one reason why gold nanoparticles can appear a different colour and show different chemical reactivity from bulk gold metal. IB-CHEMISTRY-S2.4-4 PRACTICE
Chemistry HL/SL S3.1 The periodic table: classification of elements State the general trend in first ionization energy across a period and down a group, and give the reason for each. IB-CHEMISTRY-S3.1-1 CONCEPT	Chemistry HL/SL S3.1 The periodic table: classification of elements How can a large jump between successive ionization energies of an element be used to deduce its group number? IB-CHEMISTRY-S3.1-2 CONCEPT	Chemistry HL/SL S3.1 The periodic table: classification of elements Which period 3 element has the highest first ionization energy? A Cl B S C P D Si IB-CHEMISTRY-S3.1-3 MULTIPLE CHOICE
Chemistry HL/SL S3.1 The periodic table: classification of elements State whether atomic radius increases or decreases (i) across period 2 from Li to F, and (ii) down group 1 from Li to Cs, and give one reason for each. IB-CHEMISTRY-S3.1-4 PRACTICE	Chemistry HL/SL S3.2 Functional groups: classification of organic compounds What defines a homologous series, and give one example of a trend shown within a series. IB-CHEMISTRY-S3.2-1 CONCEPT	Chemistry HL/SL S3.2 Functional groups: classification of organic compounds Distinguish between chain isomerism and positional isomerism, with an example of each. IB-CHEMISTRY-S3.2-2 CONCEPT
Chemistry HL/SL S3.2 Functional groups: classification of organic compounds Which homologous series has the general formula $C_nH_{2n+1}COOH$? A Alkane B Alkene C Carboxylic acid D Amine IB-CHEMISTRY-S3.2-3 MULTIPLE CHOICE	Chemistry HL/SL S3.2 Functional groups: classification of organic compounds Name the functional group and homologous series of the compound $CH_3CH_2CH_2Cl$, and give its IUPAC name. IB-CHEMISTRY-S3.2-4 PRACTICE	Chemistry HL/SL R1.1 Measuring enthalpy changes State the equation used to calculate heat transferred in a calorimetry experiment, and define each symbol. IB-CHEMISTRY-R1.1-1 CONCEPT
Chemistry HL/SL R1.1 Measuring enthalpy changes Why is the enthalpy change measured in a simple calorimetry experiment usually less exothermic (numerically smaller) than the true/literature value? IB-CHEMISTRY-R1.1-2 CONCEPT	Chemistry HL/SL R1.1 Measuring enthalpy changes Which of the following is an example of an exothermic process? A Combustion of a fuel B Melting of ice C Dissolving of most ammonium salts in water D Photosynthesis IB-CHEMISTRY-R1.1-3 MULTIPLE CHOICE	Chemistry HL/SL R1.1 Measuring enthalpy changes 50.0 g of water is heated and its temperature rises by 12.0 °C. Calculate the heat absorbed by the water ($c = 4.18 \text{ J g}^{-1} \text{ K}^{-1}$). IB-CHEMISTRY-R1.1-4 PRACTICE



Chemistry HL/SL R1.2 Energy cycles in reactions (HL)	Chemistry HL/SL R1.2 Energy cycles in reactions (HL)	Chemistry HL/SL R1.2 Energy cycles in reactions (HL)
State Hess's law and explain why it is valid.	Why is lattice enthalpy of an ionic compound impossible to measure directly, and how is a Born-Haber cycle used instead?	Which of these Born-Haber cycle quantities generally has the most exothermic (most negative) value for an ionic compound? A Enthalpy of formation of NaCl(s) B Lattice enthalpy of NaCl(s) C First ionization energy of Na(g) D First electron affinity of Cl(g)
IB-CHEMISTRY-R1.2-1 CONCEPT	IB-CHEMISTRY-R1.2-2 CONCEPT	IB-CHEMISTRY-R1.2-3 MULTIPLE CHOICE
Chemistry HL/SL R1.2 Energy cycles in reactions (HL)	Chemistry HL/SL R1.3 Energy from fuels	Chemistry HL/SL R1.3 Energy from fuels
Using bond enthalpies ($\text{H-H} = 436 \text{ kJ mol}^{-1}$, $\text{Cl-Cl} = 242 \text{ kJ mol}^{-1}$, $\text{H-Cl} = 431 \text{ kJ mol}^{-1}$), estimate ΔH for $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow 2\text{HCl}(\text{g})$.	Define enthalpy of combustion, $\Delta H_{\text{c}}^{\circ}$.	Why might specific energy (kJ g^{-1}) be a more useful comparison than molar enthalpy of combustion (kJ mol^{-1}) when choosing a fuel for a vehicle?
IB-CHEMISTRY-R1.2-4 PRACTICE	IB-CHEMISTRY-R1.3-1 CONCEPT	IB-CHEMISTRY-R1.3-2 CONCEPT
Chemistry HL/SL R1.3 Energy from fuels	Chemistry HL/SL R1.3 Energy from fuels	Chemistry HL/SL R2.1 How much? The amount of chemical change
What additional products, besides CO_2 and H_2O , are typically formed when a hydrocarbon fuel undergoes incomplete combustion due to insufficient oxygen? A CO_2 only B CO and soot (carbon) as well as CO_2 C SO_2 only D NO_x only	1.00 g of a fuel releases 46.0 kJ of heat when burned. Calculate its specific energy, and state one reason this value might differ from the fuel's literature/theoretical enthalpy of combustion value.	Define limiting reagent and explain how to identify it in a reaction between two reactants.
IB-CHEMISTRY-R1.3-3 MULTIPLE CHOICE	IB-CHEMISTRY-R1.3-4 PRACTICE	IB-CHEMISTRY-R2.1-1 CONCEPT
Chemistry HL/SL R2.1 How much? The amount of chemical change	Chemistry HL/SL R2.1 How much? The amount of chemical change	Chemistry HL/SL R2.1 How much? The amount of chemical change
Give the equation used to calculate percentage yield, and state why percentage yield is always $\leq 100\%$ in practice.	Which equation is used directly to calculate the amount, in mol, of solute from the concentration and volume of a solution? A $n = m/M$ B $n = cV$ C $pV = nRT$ D $n = N/NA$	25.0 cm^3 of $0.200 \text{ mol dm}^{-3} \text{ H}_2\text{SO}_4$ exactly neutralises 40.0 cm^3 of NaOH solution via $\text{H}_2\text{SO}_4 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$. Calculate the concentration of the NaOH solution.
IB-CHEMISTRY-R2.1-2 CONCEPT	IB-CHEMISTRY-R2.1-3 MULTIPLE CHOICE	IB-CHEMISTRY-R2.1-4 PRACTICE
Chemistry HL/SL R2.2 How fast? The rate of chemical change	Chemistry HL/SL R2.2 How fast? The rate of chemical change	Chemistry HL/SL R2.2 How fast? The rate of chemical change
State the collision theory conditions required for a chemical reaction to occur between two particles.	Explain, using the Maxwell-Boltzmann distribution, why increasing temperature increases reaction rate more than the corresponding increase in collision frequency alone would suggest.	How does a catalyst increase the rate of a reaction? A It increases the average kinetic energy of the reactant particles B It increases the activation energy of the reaction C It provides an alternative pathway with lower activation energy D It increases the concentration of the reactants
IB-CHEMISTRY-R2.2-1 CONCEPT	IB-CHEMISTRY-R2.2-2 CONCEPT	IB-CHEMISTRY-R2.2-3 MULTIPLE CHOICE



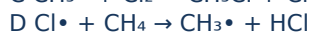
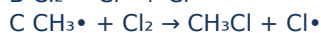
Chemistry HL/SL R2.2 How fast? The rate of chemical change	Chemistry HL/SL R2.3 How far? The extent of chemical change (HL)	Chemistry HL/SL R2.3 How far? The extent of chemical change (HL)
In an experiment, doubling [A] while keeping [B] constant causes the rate to double. What is the order of reaction with respect to A?	What characterises a system at dynamic equilibrium?	State the relationship between the standard Gibbs free energy change and the equilibrium constant, and what a large positive K_c implies about ΔG° .
IB-CHEMISTRY-R2.2-4PRACTICE	IB-CHEMISTRY-R2.3-1CONCEPT	IB-CHEMISTRY-R2.3-2CONCEPT
Chemistry HL/SL R2.3 How far? The extent of chemical change (HL)	Chemistry HL/SL R2.3 How far? The extent of chemical change (HL)	Chemistry HL/SL R3.1 Proton transfer reactions
What effect does increasing temperature have on K_c for a reaction with an endothermic forward reaction? A It has no effect on K_c B It increases K_c for an endothermic forward reaction if temperature increases C It always increases K_c regardless of ΔH D It changes K_c without changing the position of equilibrium	For the equilibrium $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$, write the K_c expression.	State the Brønsted-Lowry definitions of an acid and a base, and what defines a conjugate acid-base pair.
IB-CHEMISTRY-R2.3-3MULTIPLE CHOICE	IB-CHEMISTRY-R2.3-4PRACTICE	IB-CHEMISTRY-R3.1-1CONCEPT
Chemistry HL/SL R3.1 Proton transfer reactions	Chemistry HL/SL R3.1 Proton transfer reactions	Chemistry HL/SL R3.1 Proton transfer reactions
Explain the key chemical difference between a strong acid and a weak acid of the same concentration.	What is the pH of a $0.010 \text{ mol dm}^{-3}$ solution of a strong monoprotic acid? A 1.0 B 2.0 C 12.0 D 13.0	Calculate the pH of a $0.00100 \text{ mol dm}^{-3}$ solution of the strong base NaOH at 298 K ($K_w = 1.00 \times 10^{-14}$).
IB-CHEMISTRY-R3.1-2CONCEPT	IB-CHEMISTRY-R3.1-3MULTIPLE CHOICE	IB-CHEMISTRY-R3.1-4PRACTICE
Chemistry HL/SL R3.2 Electron transfer reactions	Chemistry HL/SL R3.2 Electron transfer reactions	Chemistry HL/SL R3.2 Electron transfer reactions
State the definitions of oxidation and reduction in terms of electron transfer and oxidation state.	In a voltaic cell, why does electron flow occur from the anode to the cathode through the external circuit, but conventional current is often described as flowing the opposite way inside the cell?	In the reaction $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaOCl} + \text{H}_2\text{O}$, chlorine undergoes disproportionation. Which statement about one part of this change is correct? A The oxidation state of Cl decreases from 0 to -1 B The oxidation state of Cl increases from -1 to 0 C Chlorine is reduced only, with no oxidation occurring D Chlorine is oxidised only, with no reduction occurring
IB-CHEMISTRY-R3.2-1CONCEPT	IB-CHEMISTRY-R3.2-2CONCEPT	IB-CHEMISTRY-R3.2-3MULTIPLE CHOICE
Chemistry HL/SL R3.2 Electron transfer reactions	Chemistry HL/SL R3.3 Electron sharing reactions	Chemistry HL/SL R3.3 Electron sharing reactions
Deduce the oxidation state of chromium in the dichromate ion, $\text{Cr}_2\text{O}_7^{2-}$.	Name and briefly describe the three stages of the free-radical substitution mechanism.	Explain, in terms of carbocation stability, why Markovnikov's rule predicts that H adds to the carbon with more hydrogens already attached in electrophilic addition to an unsymmetrical alkene.
IB-CHEMISTRY-R3.2-4PRACTICE	IB-CHEMISTRY-R3.3-1CONCEPT	IB-CHEMISTRY-R3.3-2CONCEPT



Chemistry HL/SL

R3.3 Electron sharing reactions

Which of the following represents the initiation step in the radical substitution of methane by chlorine?



IB-CHEMISTRY-R3.3-3

MULTIPLE CHOICE

Chemistry HL/SL

R3.3 Electron sharing reactions

State the major organic product formed when HCl reacts with 2-methylpropene, $(\text{CH}_3)_2\text{C}=\text{CH}_2$, via electrophilic addition, and name the type of carbocation intermediate formed.

IB-CHEMISTRY-R3.3-4

PRACTICE

Chemistry HL/SL

R3.4 Electron-pair sharing reactions (HL)

Compare the number of mechanistic steps, rate equation, and stereochemical outcome of SN1 versus SN2 substitution.

IB-CHEMISTRY-R3.4-1

CONCEPT

Chemistry HL/SL

R3.4 Electron-pair sharing reactions (HL)

Explain why primary halogenoalkanes favour the SN2 mechanism while tertiary halogenoalkanes favour SN1.

IB-CHEMISTRY-R3.4-2

CONCEPT

Chemistry HL/SL

R3.4 Electron-pair sharing reactions (HL)

Which statement correctly describes the SN2 mechanism?

A The reaction rate depends only on [nucleophile]

B The mechanism proceeds via a stable carbocation intermediate

C The nucleophile attacks from the side opposite the leaving group in a single step

D The product is always a racemic mixture

IB-CHEMISTRY-R3.4-3

MULTIPLE CHOICE

Chemistry HL/SL

R3.4 Electron-pair sharing reactions (HL)

State whether 1-bromobutane hydrolyses mainly via SN1 or SN2, and give one reason.

IB-CHEMISTRY-R3.4-4

PRACTICE



Chemistry HL/SL S1.1 Introduction to the particulate nature of matter A. Distillation vaporises and then condenses the solvent, leaving the dissolved solid behind, so the pure solvent is recovered. IB-CHEMISTRY-S1.1-3 ANSWER	Chemistry HL/SL S1.1 Introduction to the particulate nature of matter The different dye molecules have different polarities/sizes and so are retained to different extents by the stationary phase versus carried by the mobile phase, giving them different R_f values and causing them to separate. IB-CHEMISTRY-S1.1-2 ANSWER	Chemistry HL/SL S1.1 Introduction to the particulate nature of matter In a compound, different elements are chemically bonded together in a fixed ratio; in a mixture, substances are physically combined in variable ratio with no chemical bond between them. IB-CHEMISTRY-S1.1-1 ANSWER
Chemistry HL/SL S1.2 The nuclear atom The magnetic deflection depends on mass-to-charge ratio (m/z); for ions with equal charge, a larger mass gives a larger m/z and the ion is deflected less (has greater inertia against the deflecting force) than a lighter ion. IB-CHEMISTRY-S1.2-2 ANSWER	Chemistry HL/SL S1.2 The nuclear atom Atomic number (Z) = number of protons. Mass number (A) = number of protons + number of neutrons. Number of neutrons = $A - Z$. IB-CHEMISTRY-S1.2-1 ANSWER	Chemistry HL/SL S1.1 Introduction to the particulate nature of matter Melting wax: physical change — only the state changes (solid to liquid), no new substance forms. Burning the wick: chemical change — the wax/wick reacts with oxygen to form new substances (CO_2 and H_2O). IB-CHEMISTRY-S1.1-4 ANSWER
Chemistry HL/SL S1.3 Electron configurations 4s fills before 3d (Aufbau order), but when electrons are removed to form ions, 4s electrons are lost before 3d electrons, because 4s becomes higher in energy than the occupied 3d once 3d is populated. IB-CHEMISTRY-S1.3-1 ANSWER	Chemistry HL/SL S1.2 The nuclear atom $\text{Ar} = (63 \times 0.692) + (65 \times 0.308) = 43.60 + 20.02 = 63.6$ (3 s.f.), matching the periodic table value for copper. IB-CHEMISTRY-S1.2-4 ANSWER	Chemistry HL/SL S1.2 The nuclear atom B. Neutrons = mass number – atomic number = $13 - 6 = 7$. IB-CHEMISTRY-S1.2-3 ANSWER
Chemistry HL/SL S1.3 Electron configurations $1s^2 2s^2 2p^6 3s^2 3p^4$. The $3p^4$ electrons occupy the three p orbitals as $\uparrow \downarrow \uparrow \uparrow$ (Hund's rule), giving 2 unpaired electrons. IB-CHEMISTRY-S1.3-4 ANSWER	Chemistry HL/SL S1.3 Electron configurations C. Maximum electrons in shell n is $2n^2$; for $n = 3$, $2(3)^2 = 18$. IB-CHEMISTRY-S1.3-3 ANSWER	Chemistry HL/SL S1.3 Electron configurations A half-filled (Cr: $3d^5$) or fully filled (Cu: $3d^{10}$) d sub-shell is extra stable, so one 4s electron moves into the 3d sub-shell: Cr is $[\text{Ar}]4s^13d^5$ (not $4s^23d^4$) and Cu is $[\text{Ar}]4s^13d^{10}$ (not $4s^23d^9$). IB-CHEMISTRY-S1.3-2 ANSWER
Chemistry HL/SL S1.4 Counting particles by mass: the mole C. Number of particles = $n \times N_A = 0.20 \times 6.02 \times 10^{23} = 1.2 \times 10^{23}$. IB-CHEMISTRY-S1.4-3 ANSWER	Chemistry HL/SL S1.4 Counting particles by mass: the mole Experimental measurement/rounding errors in the masses of products (e.g. CO_2 and H_2O collected) mean the mole ratios calculated are rarely exact whole numbers, so they must be rounded to the nearest sensible whole-number ratio. IB-CHEMISTRY-S1.4-2 ANSWER	Chemistry HL/SL S1.4 Counting particles by mass: the mole $n = m / M$, where n is in mol, m is in grams (g), and M is the molar mass in g mol^{-1} . IB-CHEMISTRY-S1.4-1 ANSWER



Chemistry HL/SL S1.5 Ideal gases (HL)	Chemistry HL/SL S1.5 Ideal gases (HL)	Chemistry HL/SL S1.4 Counting particles by mass: the mole
At low pressure and high temperature, because the particles are far apart (so their own volume is negligible relative to the container) and moving fast (so intermolecular attractive forces have little effect), <u>matching the ideal gas assumptions.</u>	$pV = nRT$, with p in Pa, V in m^3 , n in mol, T in K ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$).	$n = c \times V = 0.100 \times 0.250 = 0.0250 \text{ mol}$. $m = n \times M = 0.0250 \times 40.0 = 1.00 \text{ g}$.
IB-CHEMISTRY-S1.5-2 ANSWER	IB-CHEMISTRY-S1.5-1 ANSWER	IB-CHEMISTRY-S1.4-4 ANSWER
Chemistry HL/SL S2.1 The ionic model	Chemistry HL/SL S1.5 Ideal gases (HL)	Chemistry HL/SL S1.5 Ideal gases (HL)
Increasing the magnitude of the ionic charges, and decreasing the ionic radii (bringing ions closer together), both increase (make more exothermic/larger) the lattice <u>enthalpy.</u>	$V = 2.00 \times 10^{-3} \text{ m}^3$. $p = nRT/V = (0.100 \times 8.31 \times 300) / (2.00 \times 10^{-3}) = 249.3 / 0.00200 = 1.25 \times 10^5 \text{ Pa} = 125 \text{ kPa}$.	$C. V = n \times V_m = 2.00 \times 22.4 = 44.8 \text{ dm}^3$.
IB-CHEMISTRY-S2.1-1 ANSWER	IB-CHEMISTRY-S1.5-4 ANSWER	IB-CHEMISTRY-S1.5-3 ANSWER
Chemistry HL/SL S2.1 The ionic model	Chemistry HL/SL S2.1 The ionic model	Chemistry HL/SL S2.1 The ionic model
Ionic compounds form a giant ionic lattice. They are hard because strong electrostatic forces act throughout the lattice in all directions, but brittle because a sharp blow shifts layers of ions so that ions of like charge become adjacent, causing strong repulsion that shatters the lattice.	A. Al_2O_3 contains Al^{3+} and O^{2-} ions with the largest ionic charges among these options, giving the strongest electrostatic attraction and hence the greatest lattice enthalpy.	In the solid, Na^+ and Cl^- ions are locked in fixed lattice positions and cannot move to carry charge; melting or dissolving frees the ions to move, so they can act as mobile charge carriers and conduct electricity.
IB-CHEMISTRY-S2.1-4 ANSWER	IB-CHEMISTRY-S2.1-3 ANSWER	IB-CHEMISTRY-S2.1-2 ANSWER
Chemistry HL/SL S2.2 The covalent model	Chemistry HL/SL S2.2 The covalent model	Chemistry HL/SL S2.2 The covalent model
B. Three electron domains (2 bonding regions + 1 lone pair) arrange trigonal planar, but with a lone pair present the molecular shape (of atoms only) is bent/angular, analogous to ozone.	Water molecules can form hydrogen bonds (O–H bonds are highly polar, and O has lone pairs) between molecules, which are much stronger than the London/van der Waals forces that are the only intermolecular force between non-polar methane molecules, so more energy is needed to separate water molecules.	The number of electron domains (bonding pairs + lone pairs) around the central atom arrange to minimise electrostatic repulsion; a lone pair repels other electron pairs more strongly than a bonding pair does, so lone pairs compress adjacent bond angles below the idealised value.
IB-CHEMISTRY-S2.2-3 ANSWER	IB-CHEMISTRY-S2.2-2 ANSWER	IB-CHEMISTRY-S2.2-1 ANSWER
Chemistry HL/SL S2.3 The metallic model	Chemistry HL/SL S2.3 The metallic model	Chemistry HL/SL S2.2 The covalent model
More delocalised electrons per atom increases the electron density in the 'sea', strengthening the overall electrostatic attraction between the cations and the delocalised electrons, so more thermal energy is required to overcome the metallic <u>bonding and melt the metal.</u>	A lattice of positive metal ions (cations) is held together by electrostatic attraction to a surrounding 'sea' of delocalised valence electrons that are free to move throughout the structure.	$\text{O}=\text{C}=\text{O}$, with 2 lone pairs on each O not directly relevant to central atom shape. Central C has 2 electron domains (double bonds count as one domain each) and no lone pairs, so CO_2 is linear with a bond angle of 180° .
IB-CHEMISTRY-S2.3-2 ANSWER	IB-CHEMISTRY-S2.3-1 ANSWER	IB-CHEMISTRY-S2.2-4 ANSWER



Chemistry HL/SL S2.4 From models to materials (HL)	Chemistry HL/SL S2.3 The metallic model	Chemistry HL/SL S2.3 The metallic model
Linear (unbranched) chains can pack closely together and align regularly, maximising the area of contact and hence the total London/dispersion forces between chains, giving a higher melting point than branched chains of the same polymer, which pack less closely.	In metals, delocalised electrons are already free to move throughout the lattice in the solid state, so solid metals conduct; in ionic solids, the charge carriers are the ions themselves, which are fixed in the lattice and only become mobile once the solid melts or dissolves.	C. Non-directional metallic bonding means cation layers can slide past one another (e.g. when drawn into a wire) while remaining held by the surrounding delocalised electrons, without breaking the structure.
IB-CHEMISTRY-S2.4-1 ANSWER	IB-CHEMISTRY-S2.3-4 ANSWER	IB-CHEMISTRY-S2.3-3 ANSWER
Chemistry HL/SL S2.4 From models to materials (HL)	Chemistry HL/SL S2.4 From models to materials (HL)	Chemistry HL/SL S2.4 From models to materials (HL)
At the nanoscale, the very high surface-area-to-volume ratio and quantum-scale dimensions change how the particles interact with light and how readily surface atoms can react, giving gold nanoparticles different optical and chemical properties from unreactive bulk gold.	A. A polyester forms by condensation between –OH and –COOH groups with loss of H ₂ O at each linkage; the other three are addition polymers formed from monomers containing a C=C double bond.	Atoms of a different size disrupt the regular arrangement of the pure metal's cation lattice, making it harder for layers of atoms to slide past each other, which increases hardness compared with the pure metal.
IB-CHEMISTRY-S2.4-4 ANSWER	IB-CHEMISTRY-S2.4-3 ANSWER	IB-CHEMISTRY-S2.4-2 ANSWER
Chemistry HL/SL S3.1 The periodic table: classification of elements	Chemistry HL/SL S3.1 The periodic table: classification of elements	Chemistry HL/SL S3.1 The periodic table: classification of elements
A. Ionization energy generally increases across period 3 due to increasing nuclear charge with constant shielding, and Cl (excluding the noble gas Ar) is the furthest right/most non-metallic of these options, giving it the highest first ionization energy.	A big jump occurs when the next electron removed comes from a full, lower (inner) shell rather than the outer shell; the number of ionization energies before the first big jump equals the number of valence electrons, which corresponds to the element's group number (for main groups).	Across a period (left to right): first ionization energy generally increases, because nuclear charge increases while shielding stays roughly constant, so outer electrons are held more tightly. Down a group: first ionization energy decreases, because additional electron shells increase distance and shielding, outweighing the increased nuclear charge.
IB-CHEMISTRY-S3.1-3 ANSWER	IB-CHEMISTRY-S3.1-2 ANSWER	IB-CHEMISTRY-S3.1-1 ANSWER
Chemistry HL/SL S3.2 Functional groups: classification of organic compounds	Chemistry HL/SL S3.2 Functional groups: classification of organic compounds	Chemistry HL/SL S3.1 The periodic table: classification of elements
Chain isomerism: same functional group and molecular formula but a different arrangement of the carbon skeleton, e.g. butane vs 2-methylpropane. Positional isomerism: same carbon skeleton and functional group but the functional group is attached at a different position, e.g. butan-1-ol vs butan-2-ol.	A homologous series is a family of compounds with the same general formula and functional group, differing by a CH ₂ unit between consecutive members, showing similar chemical properties and a gradual trend in physical properties, e.g. boiling point increases steadily with increasing chain length (more London forces).	(i) Decreases across period 2: nuclear charge increases while the number of shells/shielding stays the same, so outer electrons are pulled closer to the nucleus. (ii) Increases down group 1: each element adds an extra occupied shell, and the greater distance/extra shielding outweighs the increased nuclear charge.
IB-CHEMISTRY-S3.2-2 ANSWER	IB-CHEMISTRY-S3.2-1 ANSWER	IB-CHEMISTRY-S3.1-4 ANSWER
Chemistry HL/SL R1.1 Measuring enthalpy changes	Chemistry HL/SL S3.2 Functional groups: classification of organic compounds	Chemistry HL/SL S3.2 Functional groups: classification of organic compounds
$q = mc\Delta T$, where q is heat transferred (J), m is the mass of the substance absorbing/releasing the heat (g), c is its specific heat capacity ($\text{J g}^{-1} \text{K}^{-1}$), and ΔT is the temperature change (K or °C).	It contains a halogenoalkane functional group (–Cl); it belongs to the halogenoalkane homologous series; IUPAC name: 1-chloropropane.	C. Carboxylic acids have general formula $\text{C}_n\text{H}_{2n+1}\text{COOH}$, containing the –COOH functional group.
IB-CHEMISTRY-R1.1-1 ANSWER	IB-CHEMISTRY-S3.2-4 ANSWER	IB-CHEMISTRY-S3.2-3 ANSWER



Chemistry HL/SL R1.1 Measuring enthalpy changes	Chemistry HL/SL R1.1 Measuring enthalpy changes	Chemistry HL/SL R1.1 Measuring enthalpy changes
$q = mc\Delta T = 50.0 \times 4.18 \times 12.0 = 2508 \text{ J} \approx 2.51 \text{ kJ}.$	A. Combustion reactions release energy to the surroundings as heat and light, so they are exothermic; the others listed are typically endothermic.	Heat is lost to the surroundings (container, air) rather than all of it being transferred to/measured in the water/solution, so the measured temperature change — and hence the calculated $ \Delta H $ — is smaller than the true value; poor insulation and evaporation also contribute to heat loss.
IB-CHEMISTRY-R1.1-4 ANSWER	IB-CHEMISTRY-R1.1-3 ANSWER	IB-CHEMISTRY-R1.1-2 ANSWER
Chemistry HL/SL R1.2 Energy cycles in reactions (HL)	Chemistry HL/SL R1.2 Energy cycles in reactions (HL)	Chemistry HL/SL R1.2 Energy cycles in reactions (HL)
B. Lattice enthalpy (formation, from gaseous ions to solid lattice) is typically the largest exothermic term, since it results from strong electrostatic attraction bringing many oppositely-charged ions together into a stable solid lattice.	Lattice enthalpy corresponds to a hypothetical, non-achievable single step (gaseous ions directly forming/separating from a solid lattice) that cannot be carried out or measured experimentally in isolation. A Born–Haber cycle combines it with other measurable enthalpy changes (atomisation, ionization energy, electron affinity, enthalpy of formation) in a closed cycle, so it can be calculated indirectly using Hess's law (ΔH values around the cycle sum to zero, or the enthalpy of formation route equals the alternative route).	Hess's law states that the total enthalpy change for a reaction is the same regardless of the route taken from reactants to products. It is valid because enthalpy is a state function — it depends only on the initial and final states of the system, not on the path taken.
IB-CHEMISTRY-R1.2-3 ANSWER	IB-CHEMISTRY-R1.2-2 ANSWER	IB-CHEMISTRY-R1.2-1 ANSWER
Chemistry HL/SL R1.3 Energy from fuels	Chemistry HL/SL R1.3 Energy from fuels	Chemistry HL/SL R1.2 Energy cycles in reactions (HL)
Vehicles carry a limited mass/volume of fuel, not a fixed number of moles, so comparing energy released per gram (or per unit volume) directly indicates which fuel provides the most usable energy for the mass/space it occupies.	The enthalpy change when one mole of a substance is completely burned in excess oxygen under standard conditions (100 kPa, usually 298 K), with all reactants and products in their standard states.	Bonds broken: $436 + 242 = 678 \text{ kJ}.$ Bonds formed: $2 \times 431 = 862 \text{ kJ}.$ $\Delta H = 678 - 862 = -184 \text{ kJ mol}^{-1}.$
IB-CHEMISTRY-R1.3-2 ANSWER	IB-CHEMISTRY-R1.3-1 ANSWER	IB-CHEMISTRY-R1.2-4 ANSWER
Chemistry HL/SL R2.1 How much? The amount of chemical change	Chemistry HL/SL R1.3 Energy from fuels	Chemistry HL/SL R1.3 Energy from fuels
The limiting reagent is the reactant that is completely consumed first, limiting the amount of product formed. To identify it, calculate the moles of each reactant available, divide by its stoichiometric coefficient in the balanced equation, and the reactant giving the smaller resulting value is the limiting reagent (present in the smaller amount relative to what the equation requires).	Specific energy = $46.0 \text{ kJ} / 1.00 \text{ g} = 46.0 \text{ kJ g}^{-1}.$ It may differ from the literature value because of heat loss to the surroundings during the experiment (incomplete transfer of heat to the measuring system), or incomplete combustion of the fuel.	B. Incomplete combustion (limited O_2) produces carbon monoxide (CO) and/or unburned carbon (soot) in addition to some CO_2 and $\text{H}_2\text{O}.$
IB-CHEMISTRY-R2.1-1 ANSWER	IB-CHEMISTRY-R1.3-4 ANSWER	IB-CHEMISTRY-R1.3-3 ANSWER



Chemistry HL/SL R2.1 How much? The amount of chemical change $n(\text{H}_2\text{SO}_4) = 0.200 \times (25.0/1000) = 0.00500 \text{ mol}$. $n(\text{NaOH}) = 2 \times 0.00500 = 0.0100 \text{ mol}$. $c(\text{NaOH}) = 0.0100 / (40.0/1000) = 0.250 \text{ mol dm}^{-3}$. IB-CHEMISTRY-R2.1-4ANSWER	Chemistry HL/SL R2.1 How much? The amount of chemical change B. $n = cV$, where c is concentration (mol dm^{-3}) and V is volume (dm^3), gives the amount of solute in mol. IB-CHEMISTRY-R2.1-3ANSWER	Chemistry HL/SL R2.1 How much? The amount of chemical change % yield = (actual yield / theoretical yield) $\times 100$. It is always $\leq 100\%$ because of practical losses such as incomplete reaction, competing side reactions, and product lost during handling, transfer or purification. IB-CHEMISTRY-R2.1-2ANSWER
Chemistry HL/SL R2.2 How fast? The rate of chemical change C. A catalyst provides an alternative reaction mechanism/pathway with a lower activation energy, increasing the proportion of successful collisions, without itself being consumed. IB-CHEMISTRY-R2.2-3ANSWER	Chemistry HL/SL R2.2 How fast? The rate of chemical change Raising the temperature shifts and broadens the Maxwell-Boltzmann distribution towards higher energies, substantially increasing the fraction of particles with energy $\geq E_a$ (the area under the curve beyond E_a grows disproportionately), so the increase in the frequency of successful collisions is much greater than the modest increase in total collision frequency from faster-moving particles. IB-CHEMISTRY-R2.2-2ANSWER	Chemistry HL/SL R2.2 How fast? The rate of chemical change The particles must collide with (i) energy equal to or greater than the activation energy, and (ii) the correct/favourable orientation for the necessary bonds to break and reform. IB-CHEMISTRY-R2.2-1ANSWER
Chemistry HL/SL R2.3 How far? The extent of chemical change (HL) $\Delta G^\circ = -RT \ln K$. A large $K_c (>>1)$ gives a large positive $\ln K$, so ΔG° is large and negative, meaning the forward reaction is thermodynamically favourable/spontaneous under standard conditions. IB-CHEMISTRY-R2.3-2ANSWER	Chemistry HL/SL R2.3 How far? The extent of chemical change (HL) The forward and reverse reaction rates are equal, and the macroscopic concentrations of all reactants and products remain constant over time, even though the forward and reverse reactions continue to occur at the particle level. IB-CHEMISTRY-R2.3-1ANSWER	Chemistry HL/SL R2.2 How fast? The rate of chemical change First order with respect to A (rate $\propto [\text{A}]^1$), since doubling $[\text{A}]$ causes a proportional ($\times 2$) increase in rate. IB-CHEMISTRY-R2.2-4ANSWER
Chemistry HL/SL R3.1 Proton transfer reactions Acid: a proton (H^+) donor. Base: a proton (H^+) acceptor. A conjugate acid-base pair consists of two species that differ from each other by exactly one H^+ (e.g. HA and A^-). IB-CHEMISTRY-R3.1-1ANSWER	Chemistry HL/SL R2.3 How far? The extent of chemical change (HL) $K_c = [\text{SO}_3]^2 / ([\text{SO}_2]^2[\text{O}_2])$. IB-CHEMISTRY-R2.3-4ANSWER	Chemistry HL/SL R2.3 How far? The extent of chemical change (HL) B. Increasing temperature favours the endothermic direction (Le Chatelier), shifting equilibrium towards products for an endothermic forward reaction, which increases K_c . IB-CHEMISTRY-R2.3-3ANSWER
Chemistry HL/SL R3.1 Proton transfer reactions $[\text{OH}^-] = 0.00100 \text{ mol dm}^{-3}$. $\text{pOH} = -\log_{10}(0.00100) = 3.00$. $\text{pH} = 14.00 - 3.00 = 11.00$. IB-CHEMISTRY-R3.1-4ANSWER	Chemistry HL/SL R3.1 Proton transfer reactions B. $[\text{H}^+] = 0.010 \text{ mol dm}^{-3}$, so $\text{pH} = -\log_{10}(0.010) = 2.0$. IB-CHEMISTRY-R3.1-3ANSWER	Chemistry HL/SL R3.1 Proton transfer reactions A strong acid dissociates essentially completely into ions in water, giving $[\text{H}^+]$ approximately equal to the initial acid concentration; a weak acid only partially dissociates, establishing an equilibrium between undissociated acid and its ions, so $[\text{H}^+]$ is much lower than the initial concentration for the same nominal concentration. IB-CHEMISTRY-R3.1-2ANSWER



Chemistry HL/SL R3.2 Electron transfer reactions	Chemistry HL/SL R3.2 Electron transfer reactions	Chemistry HL/SL R3.2 Electron transfer reactions
A. Some Cl atoms are reduced, going from oxidation state 0 (in Cl ₂) to -1 (in Cl ⁻ of NaCl); other Cl atoms are simultaneously oxidised to +1 (in OCl ⁻), which is disproportionation.	Electrons are released by oxidation at the anode and are consumed by reduction at the cathode, so they physically flow anode → cathode through the wire; ions move through the internal solution/salt bridge to balance charge, which is a separate (ionic) part of the circuit from the electron flow in the wire.	Oxidation is the loss of electrons, corresponding to an increase in oxidation state; reduction is the gain of electrons, corresponding to a decrease in oxidation state.
IB-CHEMISTRY-R3.2-3 ANSWER	IB-CHEMISTRY-R3.2-2 ANSWER	IB-CHEMISTRY-R3.2-1 ANSWER
Chemistry HL/SL R3.3 Electron sharing reactions	Chemistry HL/SL R3.3 Electron sharing reactions	Chemistry HL/SL R3.2 Electron transfer reactions
Adding H ⁺ to the carbon with more existing H atoms leaves the positive charge on the more substituted carbon, forming a more stable carbocation (secondary/tertiary carbocations are stabilised by the electron-donating/inductive effect of alkyl groups more than a primary carbocation), so this pathway is favoured and gives the major product.	Initiation: UV light homolytically splits a molecule (e.g. Cl ₂) into two radicals. Propagation: radicals react with other molecules, regenerating a new radical each time, sustaining the chain. Termination: two radicals combine to form a stable (non-radical) molecule, ending that chain.	Let oxidation state of Cr = x. O is -2 (×7 = -14). Overall charge = -2: 2x + (-14) = -2, so 2x = 12, x = +6. Chromium is in the +6 oxidation state.
IB-CHEMISTRY-R3.3-2 ANSWER	IB-CHEMISTRY-R3.3-1 ANSWER	IB-CHEMISTRY-R3.2-4 ANSWER
Chemistry HL/SL R3.4 Electron-pair sharing reactions (HL)	Chemistry HL/SL R3.3 Electron sharing reactions	Chemistry HL/SL R3.3 Electron sharing reactions
SN1: two steps via a carbocation intermediate; rate = k[halogenoalkane] only; gives a racemic mixture (nucleophile attacks the planar carbocation from either face). SN2: one concerted step with backside attack; rate = k[halogenoalkane][nucleophile]; gives inversion of configuration at the carbon.	Major product: 2-chloro-2-methylpropane, (CH ₃) ₃ CCl, formed via the more stable tertiary carbocation intermediate, (CH ₃) ₃ C ⁺ .	B. Initiation is the homolytic breaking of the Cl-Cl bond by UV light to form two chlorine radicals, Cl ₂ → Cl• + Cl•.
IB-CHEMISTRY-R3.4-1 ANSWER	IB-CHEMISTRY-R3.3-4 ANSWER	IB-CHEMISTRY-R3.3-3 ANSWER
Chemistry HL/SL R3.4 Electron-pair sharing reactions (HL)	Chemistry HL/SL R3.4 Electron-pair sharing reactions (HL)	Chemistry HL/SL R3.4 Electron-pair sharing reactions (HL)
SN2. As a primary halogenoalkane, the carbon bearing the bromine is not sterically hindered, allowing effective backside nucleophilic attack, and a primary carbocation (needed for SN1) would be too unstable to form readily.	C. SN2 is a single concerted step in which the nucleophile attacks from directly opposite the leaving halide (backside attack), causing inversion of configuration at that carbon.	Primary halogenoalkanes have little steric hindrance around the carbon bearing the halogen, allowing effective backside nucleophilic attack (SN2), and would form an unstable primary carbocation if ionised (disfavouring SN1). Tertiary halogenoalkanes are too sterically hindered for effective backside attack (disfavouring SN2), but readily form a relatively stable tertiary carbocation, favouring SN1.
IB-CHEMISTRY-R3.4-4 ANSWER	IB-CHEMISTRY-R3.4-3 ANSWER	IB-CHEMISTRY-R3.4-2 ANSWER